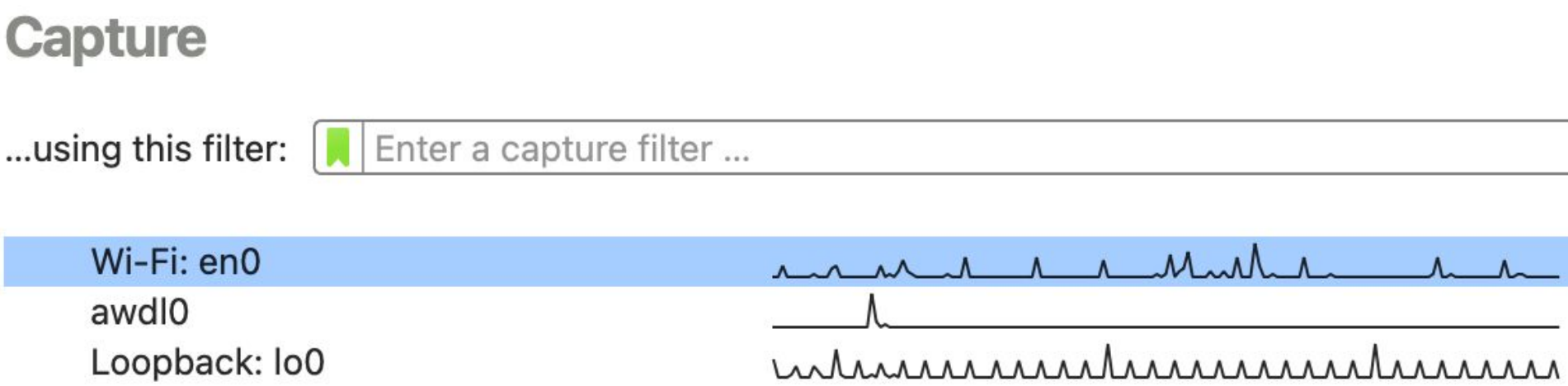
Wireshark:
Network Traffic Analysis

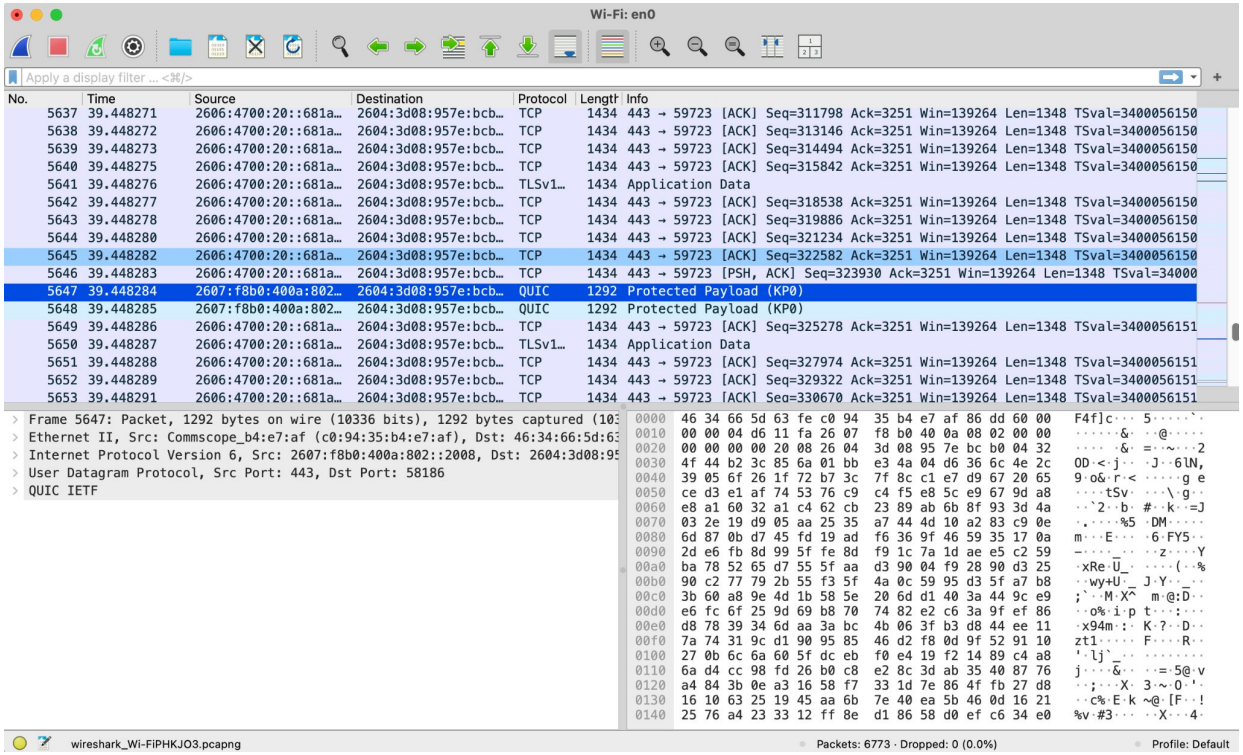
Shinyoung You

Start packet sniffing



Since I’m connected with my home wifi, I click on Wi-Fi: en0,
to be able to start packet sniffing

Wireshark sniffs all packet communications



each row represents each packet

Statistics > Conversation > TCP · 8



Ethernet · 7 | IPv4 · 10 | IPv6 · 19 | TCP · 8 | UDP · 34

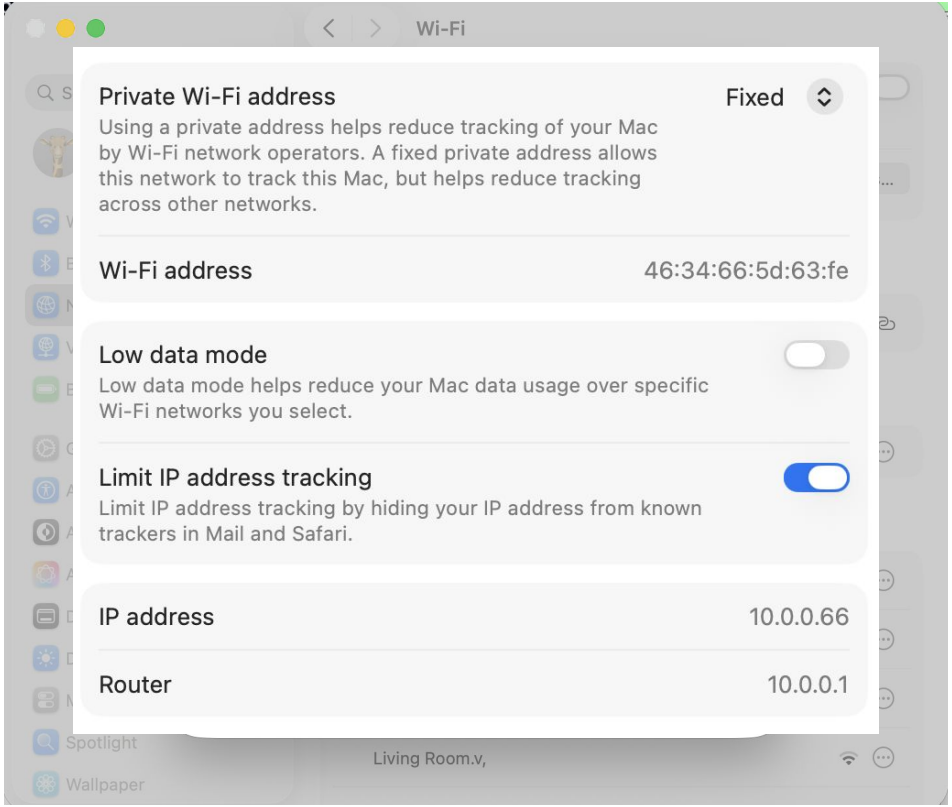
Address A	Port A	Address B	Port B	Packets	Bytes	Stream ID
140.82.113.26	443	10.0.0.66	59691	4	320 bytes	3
2602:f79a::1	443	2604:3d08:957e:bcb0:432:4f44:b23c:856a	59618	4	398 bytes	4
2604:3d08:957e:bcb0:432:4f44:b23c:856a	59838	2606:4700::6810:5049	443	9	2 kB	5
2604:3d08:957e:bcb0:432:4f44:b23c:856a	64839	2620:149:a43:380::8	443	1	74 bytes	2
2604:3d08:957e:bcb0:fc1c:6b55:7bf2:77fe	59839	2001:4e8:0:b002::18f4:1228	443	2	160 bytes	7
2604:3d08:957e:bcb0:fc1c:6b55:7bf2:77fe	59840	2606:4700::6810:5049	443	11	3 kB	6
2607:f8b0:400a:803::2003	443	2604:3d08:957e:bcb0:432:4f44:b23c:856a	59855	13	6 kB	1
2620:0:890::100	443	2604:3d08:957e:bcb0:432:4f44:b23c:856a	59831	4	356 bytes	0

Me:
- 10.0.0.66: My IP address
- 2604:3d08:957e:bcb0:432:4f44:b23c:856a: Shaw, Vancouver 1
- 2604:3d08:957e:bcb0:fc1c:6b55:7bf2:77fe: Shaw, Vancouver 2

Google LLC
- 2607:f8b0:400a:803::2003
- 2620:0:890::100

IPv6 addresses show because I'm using Wi-Fi

My Wi-Fi address

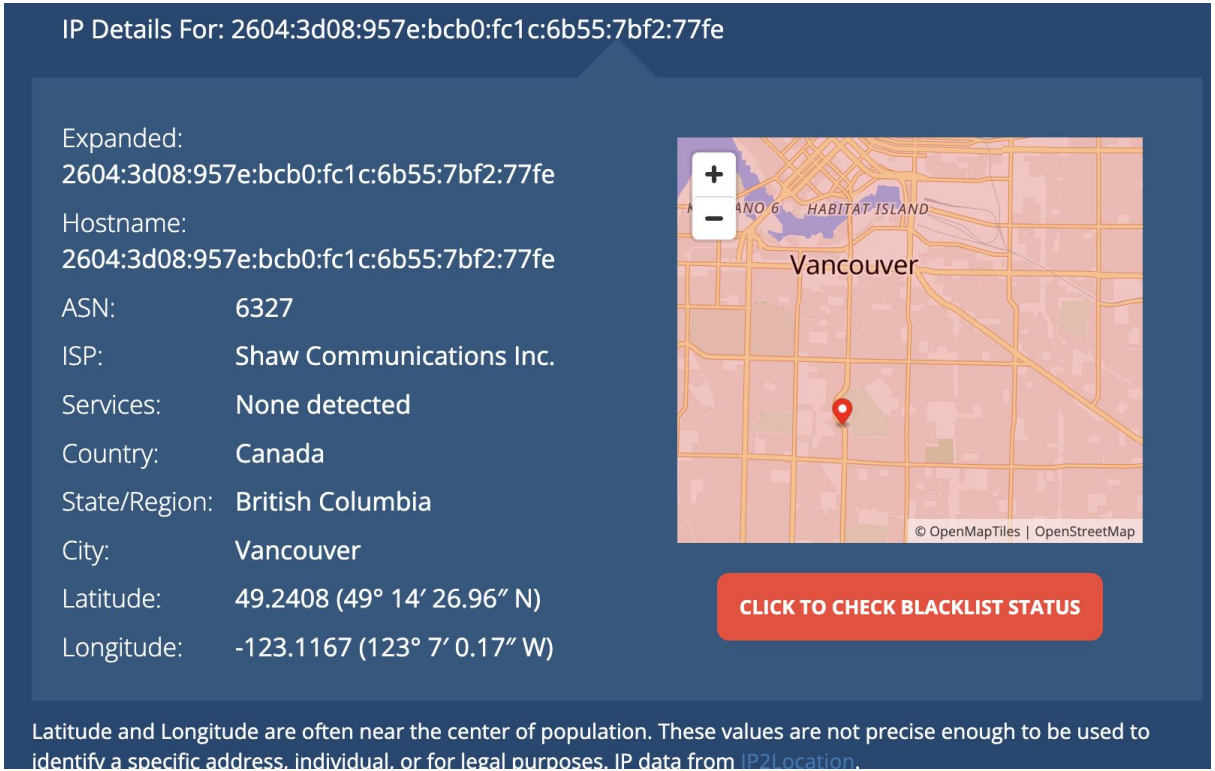


Wi-Fi address: 46:34:66:5d:63:fe

IP address: 10.0.0.66

Router: 10.0.0.1

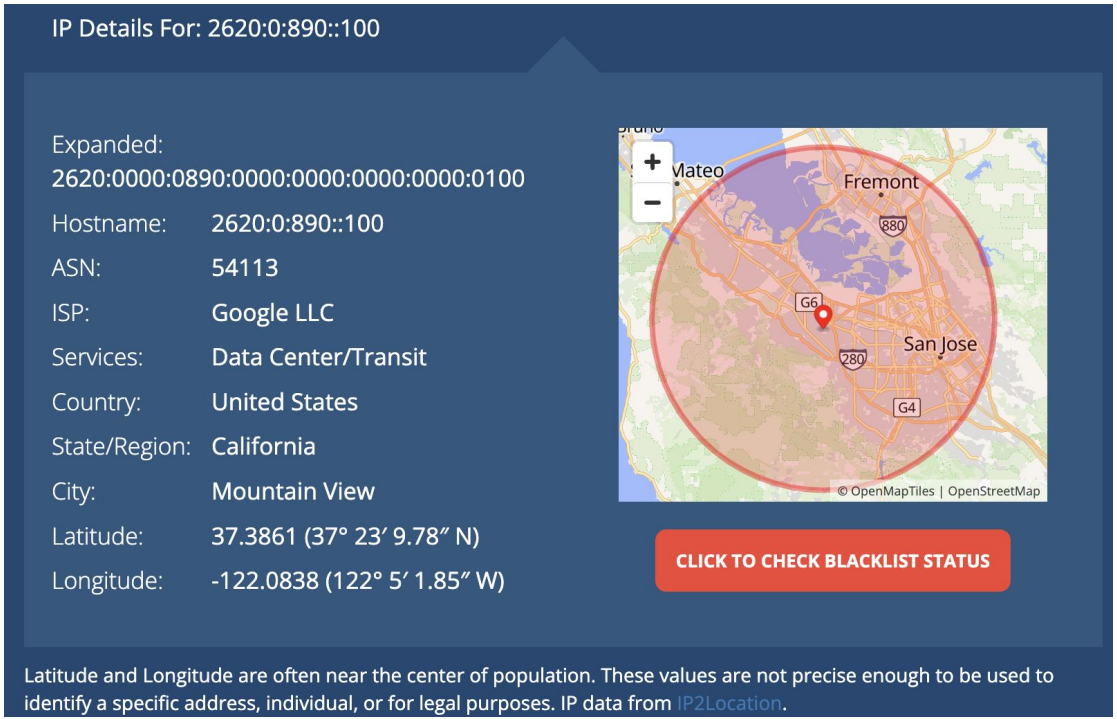
WhatIsMyIPAddress.com



Me
-2604:3d08:957e:bcb0:432:4f44:b23c:856a
-2604:3d08:957e:bcb0:fc1c:6b55:7bf2:77fe

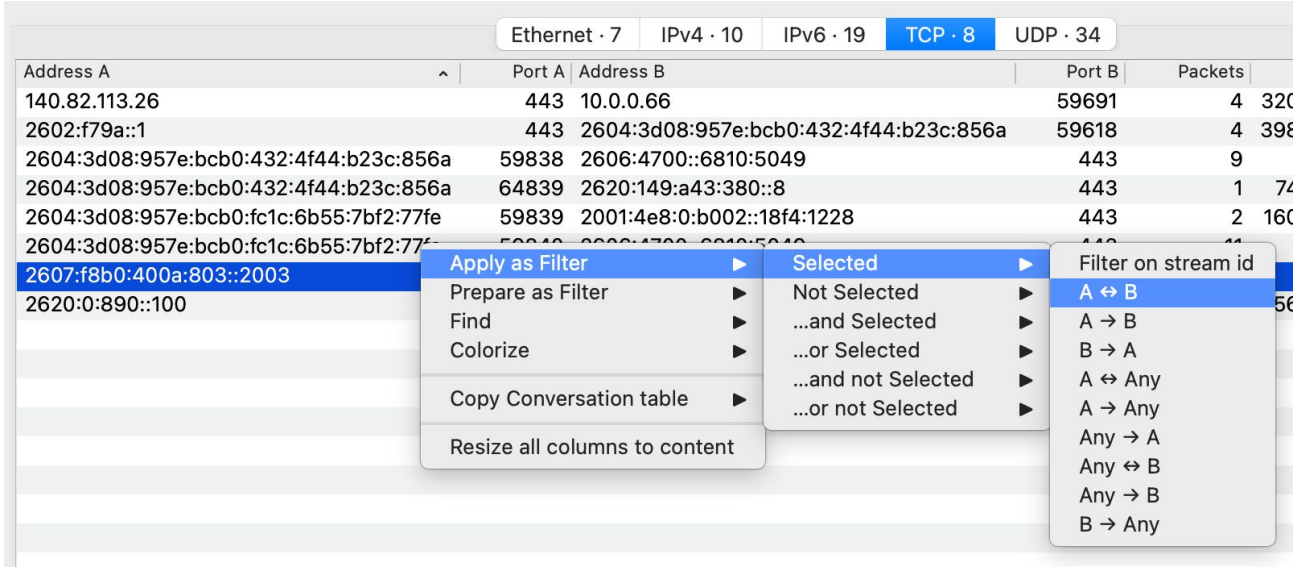
IPv6 addresses show because I’m using Wi-Fi

WhatIsMyIPAddress.com

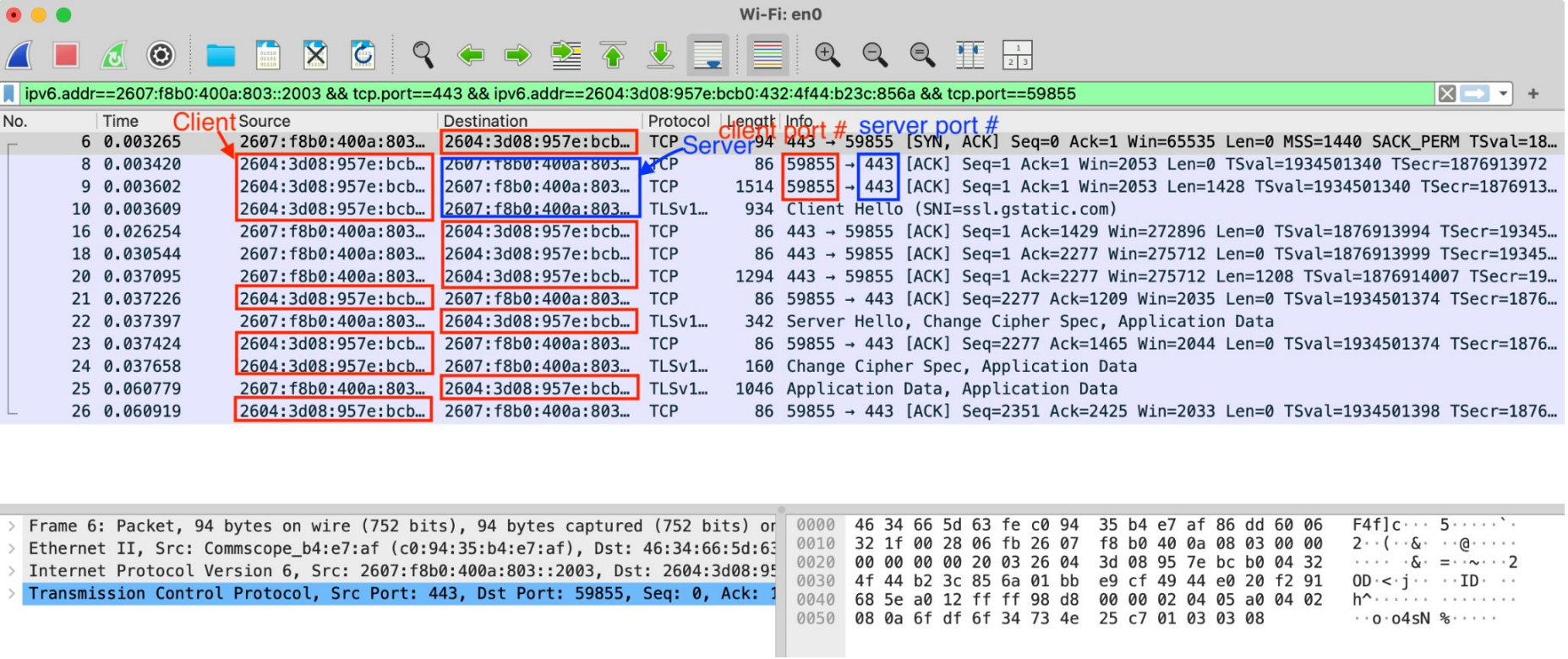


Google
- 2607:f8b0:400a:803::2003
- 2620:0:890::100

I’m communicating with Notion, GitHub, CloudFlare, and Google.
But, I only want to see the conversation between me and Google.



Apply as Filter
→ Selected
→ A ↔ B



It gives me filtered result for communication between me and Google

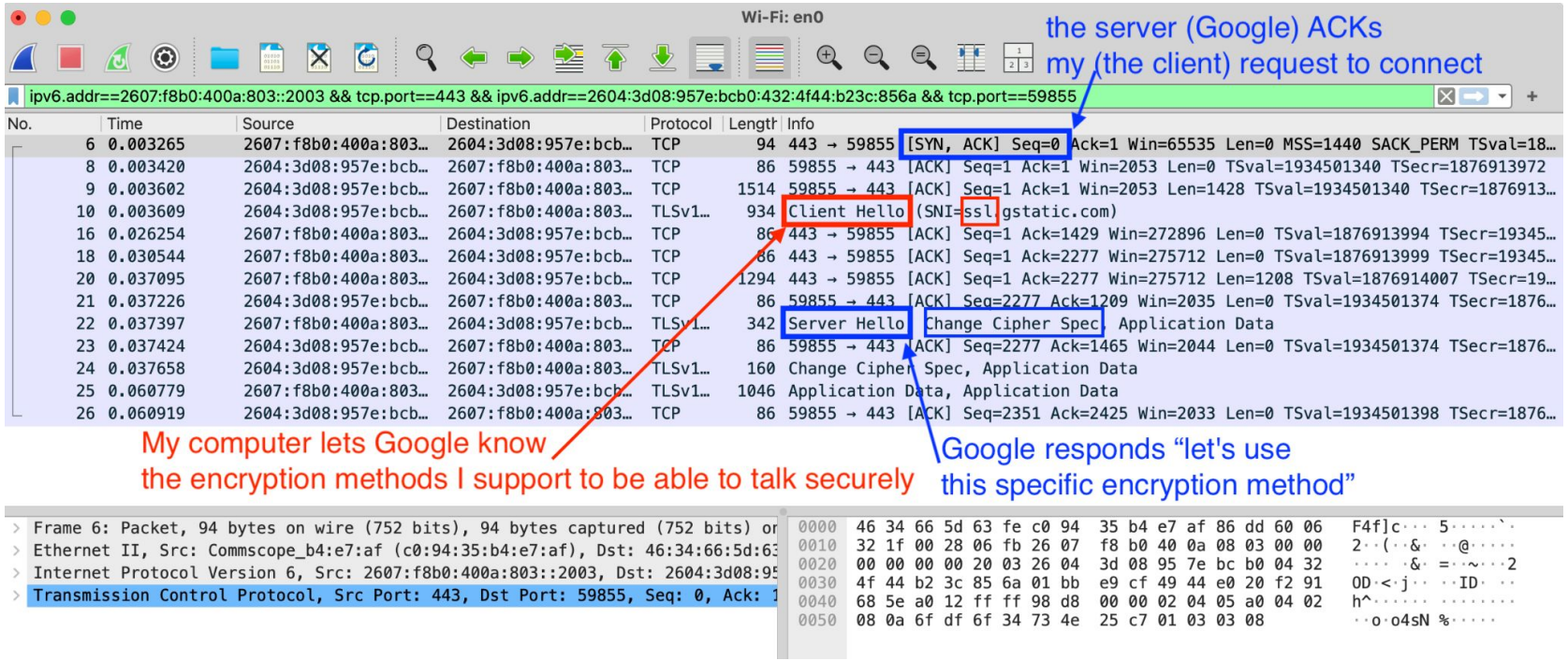
Client: Ephemeral port number (> 1023)

- My computer picks a random, temporary port number

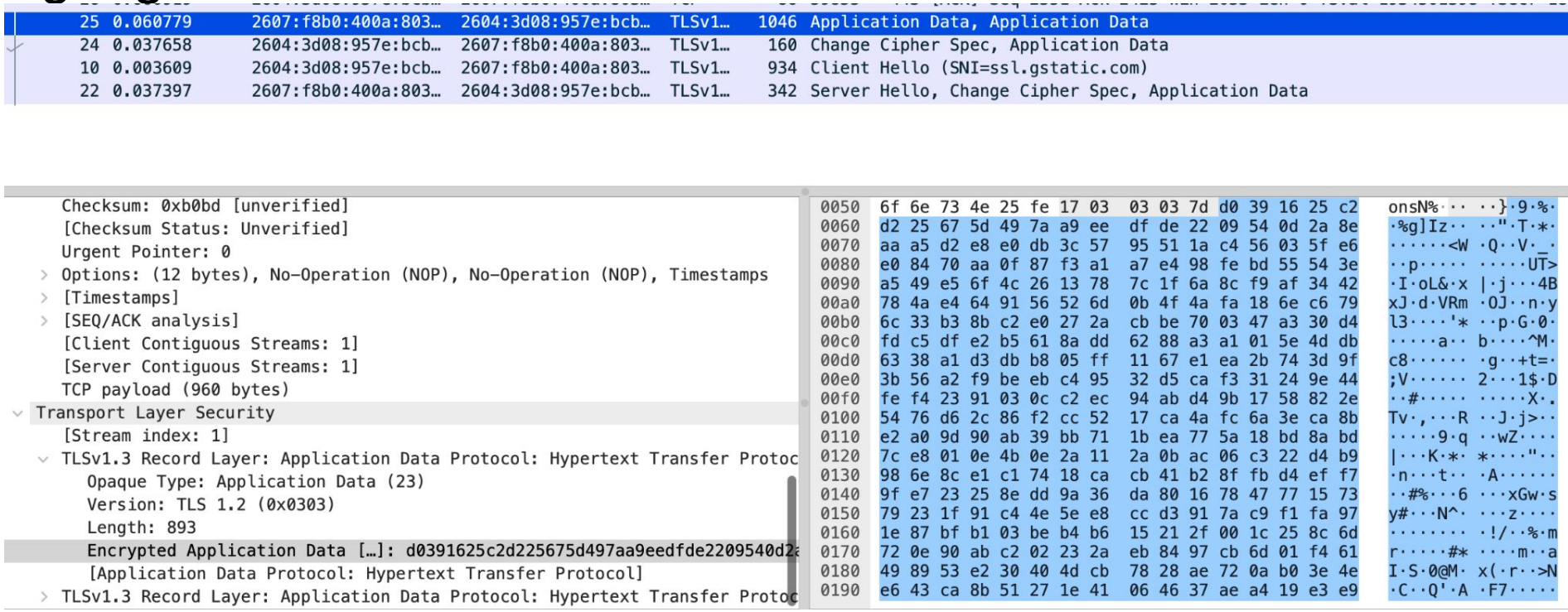
Server: Well-Known port number (≤ 1023)

- Servers need to be easy to find, so they stay at a permanent address that everyone knows

SYN & ACK + encryption method for https 443



google.com -> search for “wireshark”



The highlighted one is the message body.

I cannot read the actual message (e.g. search term) within that string because it’s encrypted as Ciphertext.

to see what is "inside" that string, I have to do the SSL Key Logging step.

Packet analysis in the OSI layer's view

```
> Frame 25: Packet, 1046 bytes on wire (8368 bits), 1046 bytes captured (8368 bits) on interface en0, id 0
> Ethernet II, Src: Commscope_b4:e7:af (c0:94:35:b4:e7:af), Dst: 46:34:66:5d:63:fe (46:34:66:5d:63:fe)
> Internet Protocol Version 6, Src: 2607:f8b0:400a:803::2003, Dst: 2604:3d08:957e:bcb0:432:4f44:b23c:856a
> Transmission Control Protocol, Src Port: 443, Dst Port: 59855, Seq: 1465, Ack: 2351, Len: 960
> Transport Layer Security
```

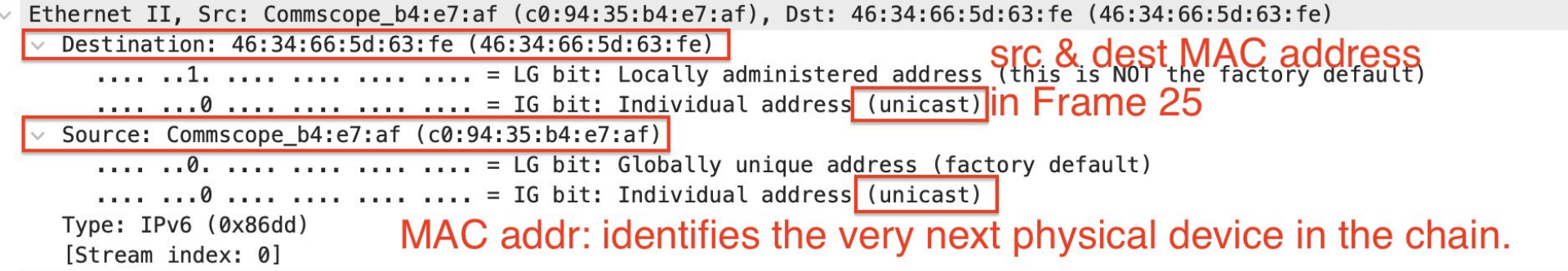
Physical layer
Data-link layer
Network layer
Transport layer
Session layer

```
Frame 25: Packet, 1046 bytes on wire (8368 bits), 1046 bytes captured (8368 bits) on interface en0, id 0
    Section number: 1
    Interface id: 0 (en0)
        Interface name: en0
        Interface description: Wi-Fi
    Encapsulation type: Ethernet (1)
    Arrival Time: May  7, 2026 14:03:50.419400000 PDT
    UTC Arrival Time: May  7, 2026 21:03:50.419400000 UTC
    Epoch Arrival Time: 1778187830.419400000
    [Time shift for this packet: 0.000000000 seconds]
    [Time delta from previous captured frame: 23.121000 milliseconds]
    [Time delta from previous displayed frame: 23.121000 milliseconds]
    [Time since reference or first frame: 60.779000 milliseconds]
    Frame Number: 25
    Frame Length: 1046 bytes (8368 bits)
    Capture Length: 1046 bytes (8368 bits)
    [Frame is marked: False]
    [Frame is ignored: False]
    [Protocols in frame: eth:ethertype:ipv6:tcp:tls]
    Character encoding: ASCII (0)
    [Coloring Rule Name: TCP]
    [Coloring Rule String: tcp]
```

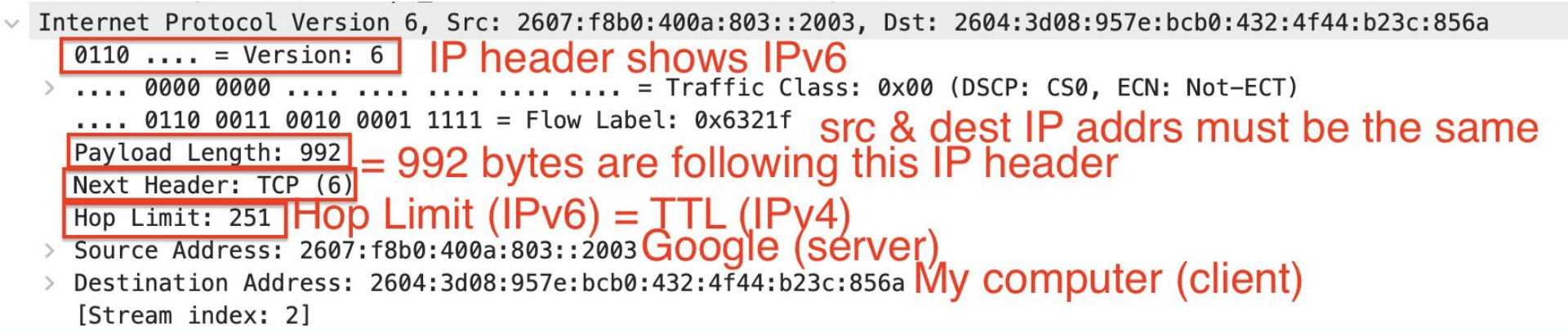
en0: physical Wi-Fi antenna or Ethernet port

Speed and timing are important in data communication

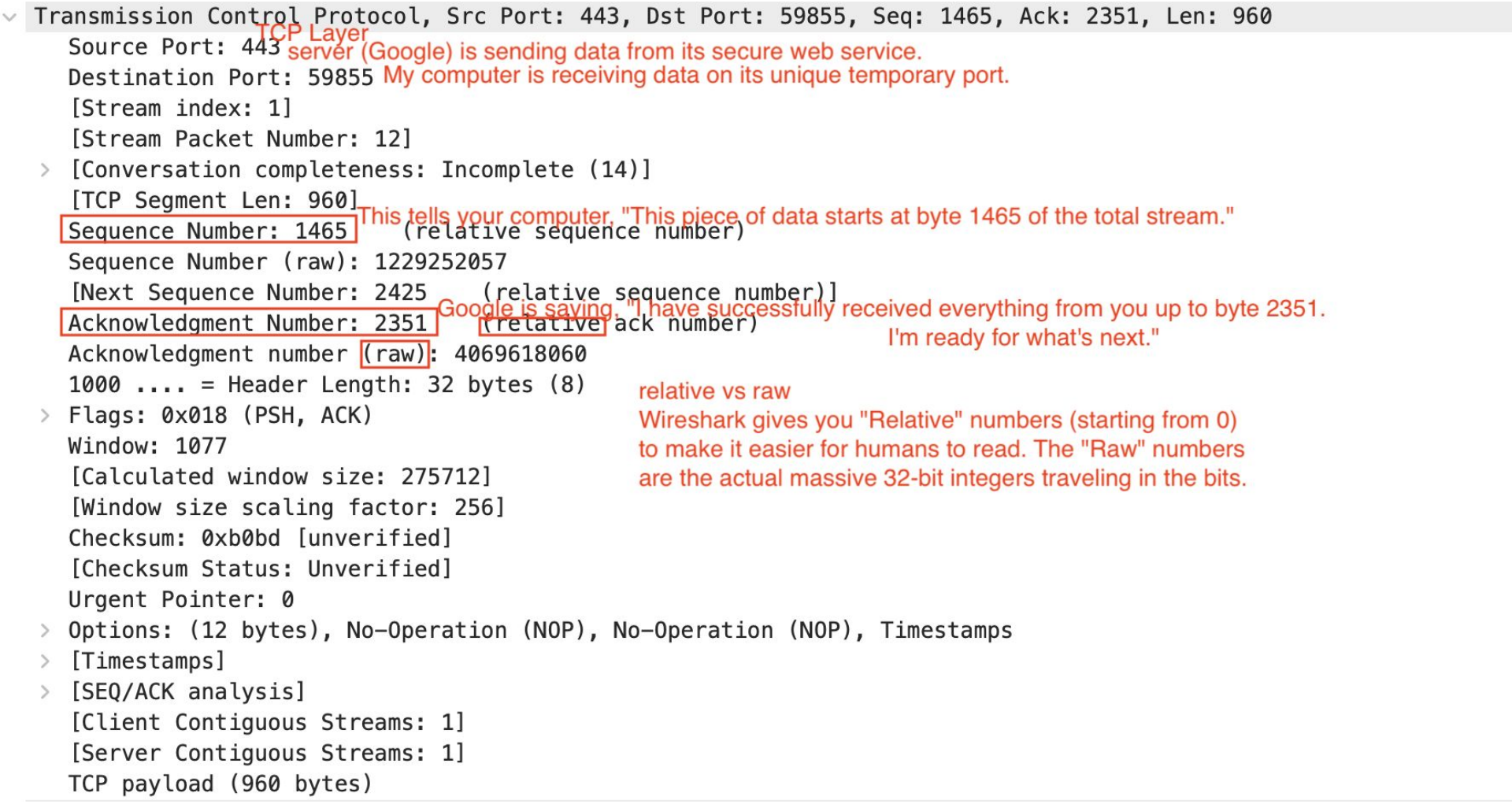
Physical layer is about bits



dest MAC addr = My Wifi address: 46:34:66:5d:63:fe



Hop Limit: Once the number hits 0, the very next router drops the packet and sends an error message (ICMPv6) back to the sender. It's a "self-destruct" timer to keep the internet clean.


Transmission Control Protocol, Src Port: 443, Dst Port: 59855, Seq: 1465, Ack: 2351, Len: 960
TCP Layer
Source Port: 443
server (Google) is sending data from its secure web service.
Destination Port: 59855
My computer is receiving data on its unique temporary port.
[Stream index: 1]
[Stream Packet Number: 12]
[Conversation completeness: Incomplete (14)]
[TCP Segment Len: 960]
Sequence Number: 1465 (relative sequence number)
This tells your computer, "This piece of data starts at byte 1465 of the total stream."
Sequence Number (raw): 1229252057
[Next Sequence Number: 2425 (relative sequence number)]
Acknowledgment Number: 2351 (relative ack number)
Google is saying, "I have successfully received everything from you up to byte 2351.
I'm ready for what's next."
Acknowledgment number (raw): 4069618060
1000 .... = Header Length: 32 bytes (8)
Flags: 0x018 (PSH, ACK)
relative vs raw
Wireshark gives you "Relative" numbers (starting from 0)
to make it easier for humans to read. The "Raw" numbers
are the actual massive 32-bit integers traveling in the bits.
Window: 1077
[Calculated window size: 275712]
[Window size scaling factor: 256]
Checksum: 0xb0bd [unverified]
[Checksum Status: Unverified]
Urgent Pointer: 0
Options: (12 bytes), No-Operation (NOP), No-Operation (NOP), Timestamps
[Timestamps]
[SEQ/ACK analysis]
[Client Contiguous Streams: 1]
[Server Contiguous Streams: 1]
TCP payload (960 bytes)

```
Transport Layer Security
  [Stream index: 1]
  TLSv1.3 Record Layer: Application Data Protocol: Hypertext Transfer Protocol
    Opaque Type: Application Data (23)
    Version: TLS 1.2 (0x0303)
    Length: 893
    Encrypted Application Data […]: d0391625c2d225675d497aa9eedfde2209540d2a8eaaa5d2e8e0db3c5795511ac456035fe6e08470aa0f
    [Application Data Protocol: Hypertext Transfer Protocol]
  TLSv1.3 Record Layer: Application Data Protocol: Hypertext Transfer Protocol
    Opaque Type: Application Data (23)
    Version: TLS 1.2 (0x0303)
    Length: 57
    Encrypted Application Data: e6c8359ca0696724792fcaed5a80bbc191974067653a3f48a86728585b1595b9b3b60c40cbdf908f718503c7
    [Application Data Protocol: Hypertext Transfer Protocol]
```